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Effect of pedestrian characteristics and their road crossing behaviors on driver yielding behavior at controlled intersections

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ABSTRACT

Introduction: Globally, pedestrians are one of the most vulnerable road-user groups. Their vulnerability increases while crossing the road at controlled intersections during the "*don't walk*" phase. Previous literature shows that driver yielding behavior has an association with pedestrian safety at intersections. Though several studies have explored driver yielding behavior towards pedestrians at conflict points, evidence on how pedestrian actions influence driver yielding behavior at intersections is yet to be investigated. *Method*: To pursue this end, a binary logistic regression model was developed using the collected data to explore the effect of non-compliant pedestrian characteristics and their road crossing behavior on driver yielding behavior towards pedestrians at six controlled intersections of Dhaka, Bangladesh. The data were collected through videography survey. *Results:* Results showed that drivers were more likely to yield to pedestrians who were female, crossing in a group, carrying baggage, not using a mobile, making some hand gesture to the driver, or crossing by rolling gap strategy. *Practical Applications:* These findings add new insights for transportation planners into the complex interaction between vehicles and pedestrians at busy controlled intersections, and thus would help to make a pedestrian friendly street.
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1. Introduction

Pedestrians are one of the most vulnerable road-user groups throughout the world (Saha, Tishi, Islam, & Mitra, 2013; Fu, Miranda-Moreno, & Saunier, 2018). According to the World Health Organization (WHO, 2018), about 1.35 million people die due to road crashes worldwide every year. Pedestrians constitute around 22% of those deaths. In addition, 93% of the road crash-related deaths occur in the low-and middle-income countries. In low-income countries like Bangladesh, pedestrian fatalities have been increasing over the years due to ineffective road safety strategies (Khan & Rahman, 2016). According to the Road Safety Facts (2013) of Bangladesh, a total of 28,244 pedestrian casualties occurred from the year of 1998 to 2013 (ARI, 2016). That represents 34% of all traffic-related casualties. About 31% of the pedestrian casualties took place when they were crossing the road. On top of everything, approximately 40% of the pedestrian casualties occurred at or near intersections. The casualty rate of the pedestrian highlights the vulnerability of the pedestrian group and suggests that pedestrian safety measures should be taken for reducing traffic casualties.

It seems likely that a large number of pedestrian trips involve road crossing at intersection. In high-income countries, pedestrians cross the road in an orderly manner following the pedestrian traffic lights at intersections. On the other hand, in low-income country's cities like Dhaka, the capital of Bangladesh, traffic signals are not functional at intersections (Rahman & Khadem, 2012). Traffic police control the traffic movement at intersections, mostly focusing on vehicular movement. Therefore, pedestrians cross the road in an irregular manner. Showing non-compliance behavior with the traffic control system is commonplace by both pedestrians and drivers. Zafri, Rony, and Adri (2020) observed that about 62% of the pedestrians showed non-compliance behavior with the traffic police direction at controlled intersections in Dhaka. Showing non-compliance behavior at controlled intersections increases inadvertent interactions between vehicles and pedestrians in almost all crossing events (Sun & Lu, 2011). This interaction is one of the potential conflict points involving the highest degree of vulnerability for pedestrians (Shankar, Ulfarsson, Pendyala, & Nebergall, 2003).

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Given the context, driver yielding behavior largely dictates the safety of the pedestrians while crossing at controlled intersections. If driver yielding behavior does not meet with the anticipation of the pedestrian, it results in an unsafe crossing. Studies showed that higher driver yielding rates were closely related to fewer pedestrian crashes. Schneider, Qin, Shaon, Sanatizadeh, He, Wkyhuis, Block, Santiago, and Bill (2017) found that driver yielding rates were higher at intersections having no record of pedestrian crash compared to those having record of at least two pedestrian crashes. According to Bangladesh Road Transport Authority (BRTA), Dhaka Metropolitan Police (DMP), and Dhaka Transport Coordination Authority (DTCA), 90% of the crashes occurred due to the rash driving behavior of the drivers with no intention to yield towards pedestrians (Farin, 2018).

Previous literature shows that driver yielding behavior toward pedestrians depends on several factors, including driver characteristics, roadway design and site factors, vehicular factors, safety treatment factors, and so on (Geruschat & Hassan, 2005; Mitman, Cooper, & Dubose, 2010; Piff, Stancato, Côté, Mendoza-Denton, & Keltner, 2012; Bertulis & Dulaski, 2014; Fitzpatrick et al., 2014; Hirun, 2016; Wang, Jiang, Wang, & Cheng, 2016; Sanatizadeh, 2017; Schneider, Sanatizadeh, Shaon, He, & Qin, 2018). Pedestrian characteristics and their road crossing behavior could also influence driver yielding behavior since interpersonal interaction between driver and pedestrian occurs at the time of crossing. However, previous studies rarely focused on this premise of knowledge. Therefore, this study aimed to investigate this unexplored knowledge: how do non-compliant pedestrian characteristics and their crossing behavior affect driver yielding behavior towards pedestrians at controlled intersections of Dhaka, Bangladesh? The outcome of the study would help the policymakers to make safer crossing for the pedestrians at controlled intersections.

2. Literature review

Driver yielding behavior has been considered an important issue for the past two decades to abate the potential collision with the pedestrian. Two types of factors were found to influence driver yielding behavior: (1) personal or top-down factors (e.g., demographic characteristics of driver, attitude of driver towards pedestrian, their understanding of the right of way, their driving skills); and (2) environmental or bottom-up factors (e.g., temporal gap size, number of traffic lanes, number of waiting pedestrians, position of pedestrian, traffic flow direction, presence (or absence) of traffic police, roadway characteristics, safety treatments) (Harrell, 1993; Sun, Ukkusuri, Benekohal, & Waller, 2003; Sanatizadeh, 2017; Schneider et al., 2018; Malenje, Zhao, Li, & Han, 2019).

Previous literature showed that the characteristics of the drivers influenced their yielding behavior. Old drivers were found to yield more for pedestrians than younger drivers (Hirun, 2016). Comparison between the educational attainment of the drivers showed that the drivers with a bachelor degree yielded more for pedestrians compared to the drivers with lower educational attainment (Hirun, 2016). Drivers having knowledge about the right of way yielded more for the pedestrians than those who were not aware of this (Hirun, 2016). In addition, drivers' understanding about the concept of unmarked crosswalks influenced their yielding behavior (Mitman et al., 2010). In this regard, educational campaigns (public awareness) for drivers were found to have a positive influence on driver yielding behavior (Schneider & Sanders, 2015). A campus-based campaign ended up reducing the non-yield behavior of the drivers from 61.42% to 55.19% (Zhang, Gawade, Lin, & McPherson, 2013).

Vehicular factors were also found to influence driver yielding behavior. Drivers in the United States who drive expensive and high-status vehicles, were found to be less interested in yielding to pedestrians (Piff et al., 2012). In addition, the speed of the vehicle and driver yielding rate towards pedestrians had an inverse relationship (Bertulis & Dulaski, 2014; Wang et al., 2016; Schneider et al., 2018). Geruschat and Hassan (2005), for a two-lane roundabout, estimated that the driver yielding rate was as high as 75% when the vehicle speed was as low as 15 miles per hour. Whereas, the yielding rate was found to be as low as 50% with an increase of five miles per hour in vehicle speed.

Road geometry and environment were observed to influence driver yielding behavior. The yielding rate was observed lower for longer pedestrian road crossing distance (Fitzpatrick et al., 2014; Sanatizadeh, 2017). Schneider et al. (2018) found that the yielding rate of driver was observed lower in the major roadway (e.g., arterial, collector road) and unmarked crosswalk than local roadway and marked crosswalk, respectively. They also explored that drivers were more likely to yield to pedestrians when motor-vehicle traffic was comparatively lower. Provision of proper signage, pedestrian crossing signs, high-visibility crosswalks with overhead warning signs, rectangular rapid flashing beacons, advance yield signs, and road markings had a positive impact on driver yielding behavior towards pedestrians (Schneider & Sanders, 2015; Sanatizadeh, 2017). Land use was found to have an impact on driver yielding behavior. Drivers' willingness to yield was found higher at the crosswalk towards pedestrians near the bus stop (Lin, Kourtellis, Wang, & Guo, 2015; Sanatizadeh, 2017). In addition, Geruschat and Hassan (2005) observed that 79% of the drivers yielded to pedestrians while they were attempting to enter into the intersection. On the other hand, drivers appeared less likely to yield when the surrounding land use was commercial, and in the locations with a history of the vehicle–pedestrian crashes (Sanatizadeh, 2017).

Demographic characteristics of pedestrians and their crossing behavior have an effect on drivers' likeliness to yield at intersection as well. Drivers were likely to yield towards old pedestrians compared to young pedestrians (Dileep, Koshy, & Sam, 2016). Pedestrians' assertiveness, especially walking briskly to the crosswalk, resulted in higher yielding, and young pedestrians were found to be more assertive compared to the older ones (Schroeder & Rouphail, 2011; Dileep et al., 2016; Schneider et al., 2018). On the other hand, drivers were more likely to yield to the pedestrians who carried a white cane or wore brighter clothes than those who only carried an umbrella (Bourquin, Emerson, & Sauerburger, 2011; Salamati, Schroeder, Geruschat, & Rouphail, 2013). The likelihood of yielding was found to be higher when pedestrian's relative standing position at the crosswalk was less than one foot from the curb (Geruschat & Hassan, 2005). From the pedestrian end, making no eye contact with the driver or not showing any hand gestures were also found to be important factors for which the drivers were not encouraged to show yielding behavior (Guéguen, Meineri, & Eyssartier, 2015; Dey & Terken, 2017; Jiang, Wang, Bengler, Guo, & Li, 2017). Zhuang and Wu (2014) examined four types of hand gestures of pedestrians. Results of the study showed that "L-bent-level" gesture (left elbow bent with hands level and palm facing left) increased driver yielding significantly. Smiling towards drivers also increased their yielding rate (Guéguen, Eyssartier, & Meineri, 2016).

From the literature, it was found that several studies were conducted to explore the effect of driver's characteristics, roadway geometry and environmental factors, and vehicular factors on driver yielding behavior. But very few studies have tried to explore the effect of pedestrian characteristics and their road crossing behavior on driver yielding behavior in detail (in particular, effect of gender, crossing stage, crossing direction, crossing pattern, crossing group size, baggage handling, mobile usage, and usage of crosswalk were not yet explored). Moreover, no study was found

addressing these factors in the context of a low-income country's city like Dhaka. So, this research aimed to explore the effect of non-compliant pedestrian characteristics and their road crossing behavior on driver yielding behavior at controlled intersections of Dhaka to reduce the knowledge gap.

3. Data and methods

3.1. Data collection

Six busy controlled intersections of Dhaka were selected to collect data for the study (Table 1). According to the DTCA, all the approach legs of the selected intersections are part of the major road network of Dhaka. Therefore, it was reasonable to consider these intersections as busy intersections. Though all the intersections had traffic signals, they were not functional. On any regular day, traffic movement at these intersections was controlled by three or four traffic police officers from 8 a.m. to 10 p.m. They guided the traffic movement through hand gestures based on their judgment considering on-site real-time visual traffic observation. Traffic police mainly showed gestures to the vehicular traffic moment. When traffic police permitted vehicular movement for a particular period on an approach leg of the intersection, this period was "*don't walk*" phase for the pedestrians. Pedestrians are not supposed to start crossing the leg during the "*don't walk*" phase. When traffic police stopped vehicular movement on a leg for a particular period, this period was "*walk*" phase for the pedestrians. Pedestrians are allowed to cross during the "*walk*" phase. So, it is reasonable to consider that crossing during the "*walk*" phase is far safer than the "*don't walk*" phase.

Among the studied intersections, there were three cross intersections, one tee intersection, and two roundabout intersections. One leg from each intersection that had a width of six-lane or more was selected for data collection since pedestrian crossing behaviors are well understood at wide crossing location (Li & Fernie, 2010). The selected legs were termed focused legs in this study. Crosswalks were found at five of the six focused legs. Medians and sidewalks were found at all the focused legs. Land use surrounding each controlled intersection was mainly commercial. Details of the intersections are presented in Table 1.

Pedestrian and vehicular movements on the focused legs were recorded through videography survey by setting a video camera (Canon EOS 70D) at a vantage point. Two hours of recording were conducted at each intersection on a normal working day under fair weather condition during the morning peak period (11:30 a.m.–12:30 p.m.) and the evening peak period (5:00 p.m.–6:00 p.m.). All the necessary data were extracted manually from the video recordings as described in Section 3.2. Training programs were arranged for the data extractors so that they could properly extract the required data. Data for 314 pedestrian-driver interactions were extracted from the six studied intersections. Using the extracted data, a database was prepared in the Statistical Package for the Social Sciences (SPSS) software.

3.2. Sample selection and data extraction

A pedestrian was considered as a sample if s/he faced a dilemma with vehicle(s) while crossing the road during *don't walk* phase at the studied intersections (i.e., when the right of way was assigned for vehicular movement by traffic police). This dilemma is a situation where, at least one vehicle was in a location that made it difficult for the pedestrian to decide whether s/he would be able to cross in front of the vehicle or not as it was dependent on the yielding behavior of the driver of that vehicle. Additionally, traffic flow at that time was continuous (a free-flow traffic condition), and vehicle speed was not hampered by anything other than the sample pedestrian. A vehicle for which a sample pedestrian faced the dilemma of whether to cross in front of it or not was considered as a sample vehicle for the corresponding sample pedestrian. Yielding behavior of the sample vehicle's driver as well as sample pedestrian characteristics and his/her road crossing behavior were extracted from the video recordings.

Driver yielding behavior was divided into two main categories: yield and no yield. The category was defined as "yield" if the sample vehicle's driver fully stopped or decreased the speed of the vehicle to create a crossing opportunity for the corresponding sample pedestrian. "No yield" was defined for the following two scenarios.

a) When a sample pedestrian decided not to cross in front of the sample vehicle because the available gap between the vehicle and the pedestrian was too short for him/her to cross safely. At that moment, the sample vehicle's driver did not reduce the speed of the vehicle to create a crossing opportunity for the sample pedestrian.
b) The available gap between the sample vehicle and the sample pedestrian was too short to cross the road safely. However, the pedestrian took risk and crossed the road in front of the vehicle through the available gap by drastically increasing the crossing speed. In that situation, the driver either did not reduce the speed of the vehicle for that pedestrian or changed the direction of driving without reducing the speed.

Data of pedestrian characteristics (gender and age) were extracted based on the visual appearance of the pedestrians. Pedestrians were categorized into four groups based on their age: Child (0–15 years), Young (15–40 years), Middle age (40–60 years), and Old (60 years and above).

Road crossing type is one of the most important pedestrian road crossing behaviors. Three road crossing types were considered in this study: crossing stage, crossing direction, and crossing pattern. Crossing stage was divided into three categories: single-stage, two-

Table 1
Details of the Selected Intersections.

Intersection name	Type of intersection	Number of legs	Name of the focused leg	Width of focused leg (m)	Crosswalk available in focused leg	Collected sample
Gulshan-2	Cross	4	Kamal Ataturk Avenue	26	Yes	54
Bangla Motor	Cross	4	Kazi Nazrul Islam Avenue	30	Yes	43
Shahbagh	Cross	3	Moulana Bhasani Road	31	Yes	51
Sonargaon-Panthapth-ETV	Roundabout	5	Kazi Nazrul Islam Avenue	26	Yes	52
Shapla Chattar	Roundabout	4	Toynbee Circular Road	28	No	54
Jasim Uddin Road Crossing	Tee	3	Jashimuddin Avenue	24	Yes	60

stage, and rolling gap. Crossing stage was categorized as single-stage if a pedestrian crossed the road continuously in one maneuver without changing his/her speed and direction of crossing, as well as did not stop at the median while crossing. If a pedestrian crossed the road continuously without changing his/her speed and direction of crossing but stopped in the median for a while, it was categorized as two-stage crossing. Crossing stage was categorized as a rolling gap if a pedestrian crossed the road discontinuously by searching appropriate gaps between vehicles, move through the suitable vehicular gaps, and adjust his/her speed and/or direction of crossing with the vehicular flow to avoid conflicts with the vehicles (Fig. 1). The main difference between single-and two-stage crossing was that in single-stage crossing, pedestrian did not stop on the median; besides, in two-stage crossing, pedestrian stopped on the median. On the other hand, rolling gap crossing is totally different from the single-and two-stage crossing. Therefore, for simplicity, it was assumed that the impact of single-and two-stage crossing on driver yielding behavior was almost similar compared to the rolling gap crossing. Thus, single-and two-stage crossing were merged and considered as one category for further analysis.

Crossing direction was divided into two categories: perpendicular and oblique. If a pedestrian crossed the road perpendicularly (straight path), the crossing direction was categorized as perpendicular. Otherwise (zig-zag manner), it was categorized as oblique crossing. Depending on whether a pedestrian crossed the intersection by walking or running, his/ her crossing pattern was categorized into two groups: walk and run.

Data of crossing group size (crossing alone or in group), baggage handling (carrying baggage such as backpack, hand bag, shopping bag, suitcase, stroller or not), and mobile usage (using mobile or not) were also extracted from each sample pedestrian. Showing hand gesture was categorized as "yes" if a pedestrian showed any gesture by using hand to the sample vehicle; otherwise, it was categorized as "no." Head orientation of the pedestrian was categorized as "towards vehicle," if a pedestrian oriented his/her face towards the sample vehicle. Otherwise, it was categorized as "not towards vehicle." Whether a sample pedestrian used crosswalk or not while crossing the intersection, this observation was divided into two categories: "yes" if the pedestrian used crosswalk at the time of crossing in front of the sample vehicle; otherwise, it was categorized as "no." Finally, pedestrian crossing speed was measured dividing the crossing distance (road width) by crossing time (waiting time on the sidewalk or median was not included in crossing time). Table 2 summarized these predictors and their category labels with their relative proportions.

Table 2
Descriptive Statistics of Selected Variables.

Variable	Drivers' yielding behavior		Total
	Yield	No yield	
Categorical variable	*n (%)*	*n (%)*	*n (%)*
Gender			
Female	68 (84.0)	13 (16.0)	81 (25.8)
Male	148 (63.5)	85 (36.5)	233 (74.2)
Age			
Children	2 (50.0)	2 (50.0)	4 (1.3)
Young	118 (67.0)	58 (33.0)	176 (56.1)
Middle	75 (70.1)	32 (29.9)	107 (34.1)
Old	21 (77.8)	6 (22.2)	27 (8.6)
Crossing stage			
Rolling gap	80 (78.4)	22 (21.6)	102 (32.5)
Single/two-stage	136 (64.2)	76 (35.8)	212 (67.5)
Crossing direction			
Perpendicular	104 (69.3)	46 (30.7)	150 (47.8)
Oblique	112 (68.3)	52 (31.7)	164 (52.2)
Crossing pattern			
Walk	158 (70.5)	66 (29.5)	224 (71.3)
Run	58 (64.4)	32 (35.6)	90 (28.7)
Crossing group size			
Alone	141 (63.2)	82 (36.8)	223 (71.0)
Group	75 (82.4)	16 (17.6)	91 (29.0)
Baggage handling			
No	126 (64.9)	68 (35.1)	194 (61.8)
Yes	90 (75.0)	30 (25.0)	120 (38.2)
Mobile usage			
No	199 (72.1)	77 (27.9)	276 (87.9)
Yes	17 (44.7)	21 (55.3)	38 (12.1)
Showing hand gesture			
No	133 (63.3)	77 (36.7)	210 (66.9)
Yes	83 (79.8)	21 (20.2)	104 (33.1)
Head orientation			
Towards vehicle	163 (73.8)	58 (26.2)	221 (70.4)
Not towards vehicle	53 (57.0)	40 (43.0)	93 (29.6)
Using crosswalk			
No	159 (70.4)	67 (29.6)	226 (72.0)
Yes	57 (64.8)	31 (35.2)	88 (28.0)
Total	216 (68.8)	98 (31.2)	314 (100.0)
Continuous variable	**Mean (S.D.)**	**Mean (S.D.)**	**Mean (S.D.)**
Crossing speed	1.19 (0.40)	1.28 (0.39)	1.22 (0.40)

3.3. Modeling approach

Driver yielding behavior at individual level aligns well with random utility theory and discrete choice models. A binary logistic regression model was developed to estimate driver yielding behavior towards pedestrians at controlled intersections. The probability

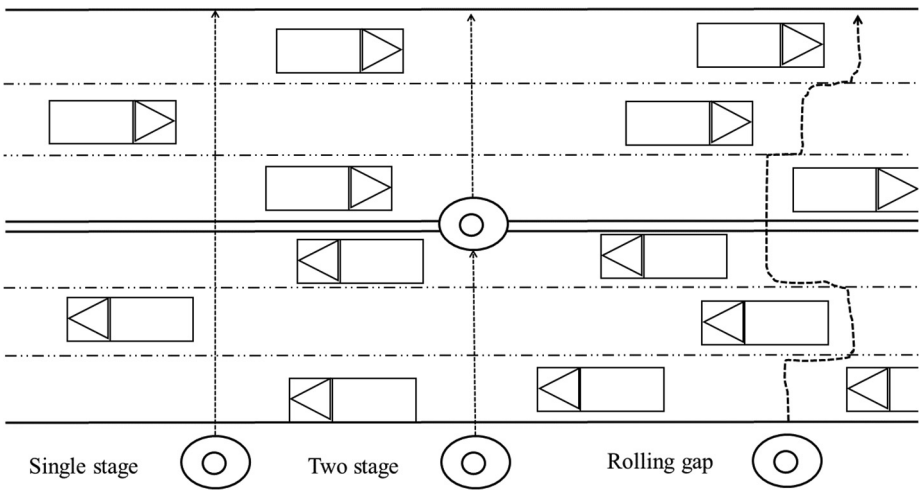


Fig. 1. Different types of crossing stage (Modified from Chandra, Rastogi, & Das, 2014).

of an individual driver to yield towards a pedestrian at intersection is given by the following formula:

$$\text{logit}(P) = ln\left(\frac{P}{1-P}\right) = \alpha_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_i x_i$$

Here, P is the probability of an individual driver to yield towards a pedestrian, α_0 is the intercept, and β_i is the model coefficient of x_i predictor.

Modeling was completed in two stages. In the first stage, predictors that should be used in developing a binary logistic regression model were identified through hypothesis testing (Chi-square ($\chi 2$) test and unpaired t test). Predictors that were found to be significant at least at 90% confidence level ($p < 0.1$) in hypothesis testing were considered for binary logistic model estimation in the second stage. After developing the model, fitness of the model with data was checked by examining the results of model statistics and Hosmer–Lemeshow test.

4. Results and discussion

4.1. Hypothesis testing

Among the collected data, in 69% of cases it was found that drivers showed yielding behavior towards pedestrians. For the purpose of modeling, hypothesis testing was conducted using the predictors listed in Table 2. A chi-square test was conducted for all the categorical predictors; whereas, unpaired t test was conducted for continuous predictor (crossing speed). Table 3 exhibits the results of the hypothesis testing. Out of all the predictors, gender, crossing stage, crossing group size, mobile usage, showing hand gesture, and head orientation appeared to be significant at 99% confidence level ($p < 0.01$). Baggage handling and crossing speed were found to be significant at 90% confidence level ($p < 0.1$). Variables that were found to be significant at least at 90% confidence level ($p < 0.1$) were considered for binary logistic model estimation stage.

4.2. Model estimation

Upon completion of the hypothesis testing, a binary logistic regression model was developed by using the eight above-mentioned significant predictors. SPSS was used to develop the model. Predictors were put into the model using the enter selection method of LOGISTIC REGRESSION in SPSS. Through this process, six predictors– gender, crossing stage, crossing group size, baggage handling, mobile usage, and showing hand gesture– of the developed model were found to be significant at 90% confidence level ($p < 0.1$) (Table 4); the likelihood of showing yielding behavior by driver was defined by the following equation:

$$\log\left[\frac{P}{1-P}\right] = 0.896 + 0.762Gen + 0.669Crs_{Stage} - 0.795Crs_{Grp} - 0.511Bag + 0.763Mobile - 0.756Hand$$

where, P = probability of showing yielding behavior by driver, Gen = Gender indicator (1 if female, 0 otherwise), Crs_{Stage}= Crossing stage indicator (1 if rolling gap, 0 otherwise), Crs_{Grp}= Crossing group size indicator (1 if alone, 0 otherwise), Bag = Baggage handling indicator (1 if no, 0 otherwise), $Mobile$ = Mobile usage indicator (1 if no, 0 otherwise), and $Hand$ = Showing hand gesture indicator (1 if no, 0 otherwise).

Chi-square statistics (χ^2), degree of freedom (df), and p-value were found 52.492, 08 and 0.000, respectively for the model (Table 4). These values indicated that the model was found to be statistically significant at 99% confidence level ($p < 0.01$). The model explained 19.1% (Nagelkerke R^2) of the variance in driver yielding behavior and correctly classified 73.3% of cases. The Hosmer and Lemeshow test result was not found to be statistically significant, $\chi^2(8) = 10.157$, p-value = 0.254 ($p > 0.05$). This result indicated that a linear relationship between the predictors and the log odds of the dependent variable was present. Therefore, the model statistics and the results of Hosmer and Lemeshow test suggested that the model fitted with the data well enough to explain driver yielding behavior towards pedestrian at controlled intersections.

Table 3
Results of Hypothesis Test.

Variable	p-value
Gender	0.000***
Age	0.568
Crossing stage	0.009***
Crossing direction	0.842
Crossing pattern	0.295
Crossing group size	0.001***
Baggage handling	0.059*
Mobile usage	0.001***
Showing hand gesture	0.002***
Head orientation	0.004***
Using crosswalk	0.341
Crossing speed	0.063*

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.3. Model results and discussion

For better understanding, the results of the model and discussion on them are presented together in this section.

4.3.1. Pedestrian characteristics related predictors

Two predictors related to pedestrian characteristics (gender and age) were considered for model estimation. Gender was only found to be significant. The result of the model showed that drivers yielded 2.1 times more towards female pedestrians compared to male pedestrians ($OR = 2.142$) indicating that the risk of female pedestrians was comparatively lower than male pedestrians (Table 4). Ferenchak (2016) found that conflicts between male pedestrians and vehicles were significantly higher than the conflicts between female pedestrians and vehicles. This finding is consistent with the result of this study as higher conflicts mean fewer exhibitions of yielding behavior by drivers towards male pedestrians and increased crash risk for them. This result could be explained from two perspectives. First, from the pedestrian's perspective, male pedestrians took more risk while crossing than female pedestrians (Granié, Abou-Dumontier, & Guého, 2013; Zafri, Rony, Rahman, & Adri, 2020). Male pedestrians might be more careless whether the vehicle yielded or not towards him than female pedestrians at the time of crossing. Secondly, from the driver's perspective, drivers yielded more towards female pedestrians might be attributed to their helpful attitude towards female pedestrians. However, further studies are required to explore the effect of pedestrians' gender on the psychology and yielding behavior of drivers. Several previous studies found that the age of the pedestrians was a significant predictor to explain driver yielding behavior (Dileep et al., 2016). However, in this study, age did not appear to be significant. It might be the presence of fewer children ($n = 4$) and old ($n = 27$) pedestrian samples.

4.3.2. Pedestrian crossing type related predictors

A broad range of pedestrian road crossing types (including crossing stage, crossing direction, and crossing pattern) were con-

Table 4
Results of the Binary Logistic Regression Model.

Variable	Coef. (*B*)	Odd ratio (*OR*)	95% *CI*		*p*-value
			Lower	Upper	
Constant	0.896	2.451			0.217
Gender (Ref: male)					
Female	0.762	2.142	1.020	4.496	0.044**
Crossing stage (Ref: single/ two-stage)					
Rolling gap	0.669	1.952	1.144	3.333	0.014**
Crossing group size (ref: group)					
Alone	−0.795	0.452	0.237	0.861	0.016**
Baggage handling (Ref: yes)					
No	−0.511	0.600	0.354	1.015	0.057*
Mobile usage (Ref: yes)					
No	0.763	2.145	1.027	4.476	0.042**
Showing hand gesture (Ref: yes)					
No	−0.756	0.470	0.262	0.843	0.011**
Head orientation (Ref: not towards vehicle)					
Towards vehicle	0.421	1.524	0.852	2.726	0.156
Crossing Speed	0.041	1.042	0.541	2.005	0.902
Model summary					
n = 314, χ^2 = 52.492, ***df*** = 08, ***p*-value** = 0.000, **−2 Log likelihood** = 391.165, **Nagelkerke *R*-square** = 0.191, **Correctly classified** = 73.3%					
Hosmer and Lemeshow test					
χ^2 = 10.157, ***df*** = 08, ***p*-value** = 0.254					

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

sidered for model estimation. Only the crossing stage was found to be significant. Drivers showed higher yielding tendency towards the pedestrians who crossed the road by rolling gap compared to the pedestrians who crossed by single-or two-stage crossing (*OR* = 1.952) (Table 4). In rolling gap crossing, a pedestrian discontinuously crosses the road by observing traffic, searching for appropriate vehicular gaps and opportunities for crossing, and boldly showing the intention to cross the road (Zafri, Sultana, Himal, & Tabassum, 2020). Shaon et al. (2018) supported this finding indirectly. They reported that pedestrian assertiveness (including standing in the road, observing the traffic, clearly expressing the intention of crossing) increased driver yielding rate by increasing visibility of the pedestrian and making the driver more aware of the presence of the pedestrian.

4.3.3. Pedestrian crossing behavior related predictors

Among the predictors related to pedestrian road crossing behavior, four factors were found to be significant; crossing group size, baggage handling, mobile usage, and showing hand gesture. Drivers were less likely to yield towards the pedestrians who crossed the road alone compared to those who crossed in a group (*OR* = 0.452) (Table 4). This result indicated that the crash risk of pedestrians who crossed in a group was lower. These findings were found consistent with previous studies (Zhuang & Wu, 2011; Ye, Wang, Cheong, & Xie, 2017). Ye et al. (2017) studied whether the pedestrian let the vehicle go first or the pedestrian went first in front of that vehicle while crossing. They found that pedestrians who crossed in groups were more likely to go first compared to pedestrians who crossed alone. In addition, Zhuang and Wu (2011) and Zafri et al. (2020) showed that the crash risk of pedestrians who crossed in a group was found lower than the pedestrians who crossed alone. In line with that, the likelihood to yield by drivers was found lower when a pedestrian did not carry any baggage while crossing compared to the pedestrian who carried baggage (*OR* = 0.600). Furthermore, group crossing behavior and carrying baggage by pedestrian might have an impact on the psychology of the drivers. Further follow up research needs to be conducted to get a clearer idea.

The probability of showing yielding behavior by drivers was found lower for the pedestrians who used a mobile while crossing than the pedestrians who did not use a mobile (*OR* = 2.145) (Table 4). This result indicates that the crash risk of the pedestrians who used a mobile while crossing was higher. A study by Nasar and Troyer (2013) also showed that using a mobile while crossing increased the chance of a pedestrian crash. This result could be described from two perspectives. First, using a mobile while crossing reduced the situational awareness and attention of pedestrian, and consequently, interpersonal communication between the approaching vehicle and inattentive pedestrian was also hampered (Nasar & Troyer, 2013; Alsaleh, Sayed, & Zaki, 2018). Since driver yielding behavior heavily depended on interpersonal communication between the driver and pedestrian, the driver yielded less for the pedestrian who communicated less with the driver due to mobile usage. The second explanation could be that mobile phone usage significantly reduced the crossing speed of pedestrians (Zhou et al., 2019). Therefore, drivers showed less yielding behavior towards them, as previous studies showed that drivers yielded less for pedestrians who crossed the road slowly (Schroeder, 2008; Schroeder & Rouphail, 2011).

In the case of showing hand gesture, drivers were less likely to yield towards the pedestrians who did not show hand gesture while crossing compared to the pedestrians who showed any gesture (*OR* = 0.470). Pedestrians who showed hand gesture towards the approaching vehicle could communicate better with the driver offering a visual cue and clear pedestrian's intention to cross (Zhuang & Wu, 2014; Dey & Terken, 2017). Therefore, the driver showed more yielding behavior towards them, and consequently, reduced the pedestrian crash risk.

5. Conclusion and recommendations

The vulnerability of pedestrians increases when conflicts between pedestrians and vehicles occur. These conflicts are more likely unavoidable if pedestrians cross the road showing non-compliance behavior at controlled intersections. However, driver yielding behavior towards pedestrians plays an important role to reduce the vulnerability of the pedestrians during conflict. Therefore, this study aimed to investigate the effect of non-compliant pedestrian characteristics and their road crossing behavior on driver yielding behavior to provide new insights into the complex interaction between pedestrians and vehicles at controlled intersections. The binary logistic regression method was employed to

model driver yielding behavior using a total of 314 pedestrian-driver interaction related data collected from six busy controlled intersections of Dhaka. The model revealed that six predictors were found to be statistically significant for explaining driver yielding behavior, including gender, crossing stage, crossing group size, baggage handling, mobile usage, and showing a hand gesture. Results of this study showed that drivers were more likely to yield towards female pedestrians than male pedestrians. It was also found that drivers yielded more towards pedestrians who crossed the road by rolling gap crossing type since pedestrians boldly showed their intention to cross in front of the vehicles while crossing by this crossing type. In addition, the likelihood of showing yielding behavior by drivers was higher for the pedestrians who crossed the road in a group, carried baggage, did not use mobile, and showed a hand gesture than the pedestrians who crossed the road alone, did not carry any baggage, used a mobile, and did not show any hand gesture while crossing, respectively.

To increase driver yielding rates, the government needs to take large scale road safety education programs periodically using leaflets, billboards, educational institute-wide training programs, social media, and mainstream media. The education programs should encourage pedestrians to cross the road in a group, to boldly express the intention of crossing in front of vehicles, and to communicate with drivers while crossing by showing visible, clear, and familiar hand gestures towards the vehicle. Therefore, pedestrians need to be educated on the importance of showing meaningful hand gestures, and drivers need to be educated to understand the gesture properly and respond to it accordingly. Mobile phone usage while crossing also needs to be highly discouraged in the education programs. Traffic regulations should be introduced to prohibit mobile phone usage during crossing. Advanced technology (e.g., installing cameras with phone usage detection technology) could help to enforce the regulations. Some education programs need to be carried out focusing on male pedestrians to reduce their vulnerability. At the time of issuing a driving license, authorities should ensure that the driver has proper knowledge about traffic laws and right of way. Effective signalized traffic control system could be designed to reduce the non-compliance behavior of the pedestrians. This would lead to reducing conflicts between drivers and pedestrians, and consequently, would improve pedestrian safety. Apart from these, the results of this study could be used to develop more reliable and precise simulation models for demonstrating driver yielding behavior. The findings would also be helpful in introducing automated vehicles in the transportation system. Finally, it can be said that the outcome of the study would further help the policymakers to take necessary interventions to alleviate pedestrian safety related problems and make a walkable and safer city for pedestrians.

Conflict of interest

The authors declare that there is no conflict of interest.

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